



AMITY UNIVERSITY
UTTAR PRADESH

AMITY SCHOOL OF ENGINEERING & TECHNOLOGY

Project Progress Report

B. Tech (Computer Science and Engineering)

Group No: 148

Project Title: Deep Learning for AI-Generated Image Detection: A Review of CNN, Vision Transformer, and Explainable Approaches

Area: Artificial Intelligence and Machine Learning (Computer Vision)

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Project Guide: Dr. Richa Gupta

Details of Project Team:

Programme:- B.TECH CSE		Year/Semester:- 4 th Year /7 th Semester	
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Paper Summary:

Introduction

Image generators such as generative adversarial networks and, more recently, diffusion models have made synthetic imagery cheap to produce and difficult to identify by eye. The practical consequences are already visible: fabricated photographic evidence, misinformation carried by realistic images, impersonation, and the contamination of image corpora scraped from the web that later become training data for other models. Manual inspection does not scale to the volume involved, so automated detection of AI-generated images has become a working requirement rather than an academic exercise.

The technical difficulty is not building a detector that works. Published detectors reach near-perfect accuracy when the images they are tested on come from the same generator they were trained on.

The difficulty is that this accuracy does not survive a change of generator. Much of what a detector learns is a consequence of one generator's design — its upsampling operator, its particular training run, its decoder — so a new generator that changes any of those does not merely present harder examples, it removes the evidence. The literature records the drop directly: a ResNet-50 reported at 99.9% accuracy on its own training generator scores 54.9% on an unseen one, and detectors reaching 92.77% mean accuracy on a standard benchmark reach 65.77% on realistic images collected in the wild. This cross-generator generalisation gap is the central open problem in the field and is the organising idea of this project.

This report covers the first phase of a two-semester project. The Semester VII deliverable is a review paper, which surveys and compares published detection methods and ends by identifying a specific gap for the second phase to fill. Studies were collected under a protocol fixed before collection began: six databases (IEEE Xplore, ScienceDirect, SpringerLink, the ACM Digital Library, arXiv and Google Scholar), eight query seeds covering GAN, diffusion and dataset-specific terms, and inclusion limited to still-image work published between 2019 and 2026 that reports quantitative results and appears either in a peer-reviewed venue or as a well-cited preprint. Video and audio deepfake detection were excluded. The searches returned 101 records; removing 26 duplicates left 75 for title-and-abstract screening, of which 74 were retained — 48 primary studies read in full and entered into a structured extraction table, 11 secondary studies used selectively, and 15 background references covering generators, backbones and explanation methods. The surveyed work is grouped below by the origin of the evidence a detector uses.

1. Handcrafted, Statistical and Frequency-Domain Forensics

Fourteen of the 48 primary studies detect generated images by measuring a property of the signal rather than by learning one end to end. The starting observation is that the repeated upsampling steps inside a generator leave periodic traces that are visible in the frequency spectrum even when nothing is visible in the picture itself. Methods in this group classify the DCT coefficients, the azimuthal average of the Fourier spectrum, co-occurrence matrices of neighbouring pixels, or a per-model fingerprint estimated as the average residual of that model's output. The reported figures are the highest anywhere in the review: 100.00% on FFHQ from DCT coefficients, above 99% from co-occurrence matrices, 99.2% on uncompressed 1024×1024 images, and 100% on a self-assembled face set from as few as 20 labelled samples.

Those figures describe narrow conditions, and the qualifying results appear alongside them almost immediately. The same spectral-decay feature that gives 99.2% at full resolution falls to 88.8% at 256×256 and becomes unusable once the image passes through 85% JPEG compression, because compression discards exactly the high frequencies the method reads. More damagingly, the traces are a consequence of a design choice rather than of synthesis itself: retraining the generators with bilinear upsampling instead of the usual operator takes detection accuracy built on those traces from 84.8–99.8% down to 0–0.3%. Later work in this group replaces hand-chosen filters with a learned frequency front-end, which is more robust in practice but inherits the same underlying dependence on how a particular generator was built.

2. CNN-Based Detection and Transfer Learning

Nineteen studies — the largest group — treat detection as ordinary binary image classification and solve it with a convolutional network pretrained on ImageNet, either frozen as a feature extractor or fine-tuned end to end. The line begins with a broad claim: a ResNet-50 trained only on ProGAN images, with careful data augmentation, averages 90.8 mAP over eleven unseen GAN architectures. Later work raises that with contrastive training, reaching 95.6% accuracy and 0.997 AUC, and with ensembles that deliberately hold a generator out of training, reaching 0.9995 AUC. Patch-level variants restrict the receptive field so that the classifier is forced onto local texture rather than global composition.

The arrival of diffusion models removed most of that margin, and the comparison table records the collapse twice. GAN-trained detectors tested on GLIDE, Latent Diffusion, Stable Diffusion, ADM and DALL-E 2 fall to roughly 60–76% accuracy on uncompressed images and to about 50–61% once those images are compressed and resized. Measured differently, the same loss is an average 15.2% AUROC drop, and it is asymmetric: a diffusion-trained detector still reaches 94.26% average detection probability on GAN images, while a GAN-trained one manages 26.34% in the reverse direction. The problem then repeats inside the diffusion family itself, where a ResNet-50 at 99.9% on its training generator scores 54.9% on another.

The response visible in the most recent rows of this group is to stop giving the classifier the image. Instead the network classifies something derived from it — the gradient map taken from a pretrained model, for 86.3% mean accuracy over eight test generators; the relationships between neighbouring pixels, for 92.2% over 28 generators; or the residual between an image and its own diffusion reconstruction, for 99.9% average accuracy over eight diffusion models. A related idea removes the trained classifier altogether and scores an image by its autoencoder reconstruction distance. One caution recurs across this group and matters for the second phase of this project: fifteen of the 48 studies train on the same ProGAN distribution, so a large share of the published cross-generator evidence measures one particular transfer rather than transfer in general.

3. Transformer and Attention-Based Approaches

Only two of the 48 primary studies belong here, which is itself a finding worth recording. A vision transformer divides an image into patches and models the relationships between them with self-attention, which suits semantic content but carries no obvious advantage when the evidence is a low-level forensic trace. The one study in the surveyed set that compares the two families under genuinely matched conditions trains a ViT-Base and an EfficientNetV2-M on the same data and tests both across fifteen manipulation methods: the CNN is better in-domain at 81.1% accuracy but falls to 47.0–57.0% on methods it has not seen, while the ViT is weaker in-domain and holds steadier at about 62.0%. The general caveat identified in this review is that backbone comparisons in this literature are usually confounded, because a change of architecture normally arrives together with a change of pretraining data and training objective, so the architecture is rarely the only variable. That observation is one of the direct reasons for the design of the second phase.

4. Multimodal and Foundation-Model-Based Detection

Eight studies build on the finding that a large vision-language model trained for something else entirely already separates real images from generated ones. With CLIP's image encoder left frozen,

a nearest-neighbour rule and a simple linear probe gain 15.07 mAP and 25.90% accuracy on unseen diffusion and autoregressive models relative to the CNN baselines of the time. The strategies built on that result read the same frozen features in different ways: a few-shot linear SVM at about 90.0% average AUC over 18 generators, intermediate encoder blocks rather than the final layer at 91.5% accuracy and 98.8 AP over 20 test sets, lightweight adapters at 98% and 95% accuracy on unseen GANs and diffusion models, prompt tuning at 98.06% mAP over 18 unseen generators, and an orthogonal subspace decomposition at 99.41% mAP and 95.19% mean accuracy. These are currently the strongest cross-generator results in the review, and their common property is that the representation itself is never trained on detection data.

5. Explainability in Detection

Explanation analysis appears in only four of the 48 rows. Grad-CAM and Grad-CAM++ are the standard tools: they weight the feature maps of a convolutional layer by the gradient of the class score and produce a heatmap of where the network looked. Applied to the small CNN released with the CIFAKE dataset, Grad-CAM shows that the classifier keys on small imperfections in image backgrounds rather than on the main subject — a useful result, because it exposes a shortcut, and a limited one, because it does not say which property of those pixels is synthetic. The remaining three rows are a patch-level map of which regions are detectable at all, a prototype-based classifier that is interpretable by construction, and a quantitative comparison of five explanation methods on a single detector. Only one of the four also carries a cross-generator test, so on the evidence of the table, explainability and generalisation are studied separately.

6. Datasets, Benchmarks and Evaluation Metrics

Eleven datasets were compared in a single table recording, for each, the introducing study, the year, the scale, the source of the real images, the generators covered and the limitation its own authors state. The line of development runs from the GAN-era ForenSynths protocol — train on ProGAN, test on eleven unseen generators — through the diffusion-era GenImage and ArtiFact collections, to in-the-wild sets such as Chameleon and TWIGMA. Three problems recur. GenImage's real and generated images differ systematically in JPEG compression and image size, so a classifier can score well by reading those properties instead of any forensic trace; removing both biases raises cross-generator accuracy by more than eleven points for both a ResNet50 and a Swin-T. A single-generator dataset such as CIFAKE cannot expose a cross-generator drop at all. And every dataset fixes its generator list at publication, so all of them go stale.

Eight evaluation metrics were then defined as numbered equations — accuracy, precision, recall, F1 score, false positive rate, AUC, average precision and mean average precision — and the review examined how the surveyed studies actually report against them. Roughly 38 quote an accuracy figure, 14 an average precision and 9 an AUC, and about 23 quote an accuracy figure and nothing else. Averages can hide the worst case: one spectrum classifier averaging 97.2% fails outright on a single generator inside its own test set. This is a direct argument for evaluating candidate architectures under one protocol rather than assembling headline figures from separate papers.

7. Comparative Analysis and the Gap Identified

All 48 primary studies were placed in a single comparison table, ordered by year, with eight columns: study, detection approach, model or architecture, dataset used, generators covered, reported accuracy or AUC, whether a cross-generator test is reported, and the limitation the authors themselves state. The last two columns are what this review adds to the conventional comparison, and every count in the analysis was obtained by tallying a column of the table so that it can be re-checked against it. The finished table was verified against the extraction records by script, and all 48 rows matched with no discrepancy in author strings, accuracy figures or cross-generator flags.

The counts give the shape of the field. Cross-generator evaluation is now common practice but not universal: 36 of the 48 studies report one and 12 do not. Twenty cover GAN-era generators only, eight diffusion-family generators only and nineteen cover both. Fifteen share the ProGAN training distribution. Ten name robustness to compression, resizing or other degradation in their own stated limitation, which makes it the most commonly acknowledged weakness in the table. Explanation analysis appears in four rows.

Reading those counts together produces the gap. Several studies do compare architectures under one protocol, and several do produce an explanation analysis, but no row of the table combines all four of the following: several ImageNet-pretrained CNN backbones trained under a single protocol, a Vision Transformer benchmark trained under the same protocol, an analysis of what the trained models attend to, and a cross-generator test. Each element is well established somewhere in the table; the four together are not reported anywhere in it. That combination is the gap this review identifies, and it defines the study designed for the project's second phase: a transfer-learning comparison of ResNet50, DenseNet121, EfficientNet-B0 and VGG19, each in a feature-extraction and a fine-tuning configuration, benchmarked against a Vision Transformer under matched conditions on CIFAKE, with Grad-CAM applied to correct and incorrect predictions and a cross-generator test on a subset of GenImage.

8. Implementation Work Begun on the Project Guide's Instruction

Following the guide's instruction of 15 August 2026 to begin preparing the implementation code ahead of Semester VIII, the experimental pipeline was built and the core experiment matrix has since been completed. None of what follows appears in the review paper, which reports no experimental results of its own; these numbers belong to the second phase of the project.

The pipeline is configuration-driven: one YAML file per experiment, and every run logs its seed, split file, optimiser, learning rate, batch size, epoch count, parameter counts, hardware string, library versions and wall-clock training time into a results CSV, so that no number is transcribed by hand into a document. CIFAKE, which holds 120,000 images at 32×32 resolution, was split once with seed 42 into 90,000 training, 10,000 validation and 20,000 test images; that split file is committed and is never regenerated, and the official test set is used only for final evaluation. Training ran on the free tier of Kaggle notebooks, and the pipeline was proved end to end on the laptop CPU before any GPU time was spent on it.

The full matrix of five backbones in two transfer-learning modes — ten runs — is now complete. Every run used the same Tesla T4 GPU, the same seed, the same split file and its committed batch size, for 19.89 hours of measured GPU time in total. Under fine-tuning the test accuracies are ViT-

B/16 98.89%, VGG19 97.91%, DenseNet121 97.60%, EfficientNet-B0 97.56% and ResNet50 95.93%. Under feature extraction they are ViT-B/16 94.75%, DenseNet121 93.49%, EfficientNet-B0 92.87%, ResNet50 92.83% and VGG19 90.55%.

Three observations are being carried into the second-phase write-up. The Vision Transformer wins in both modes, which is a direct answer to the CNN-versus-transformer question the project set out to ask. VGG19 inverts between the two modes — last under feature extraction and second under fine-tuning, with by far the largest gain of the five — which is consistent with its frozen ImageNet features encoding object semantics that transfer poorly to a task whose evidence is low-level texture. And the three fine-tuned results between 97.5% and 97.9% lie within a third of a percentage point of one another, so on a single seed they will be reported as statistically indistinguishable rather than ranked. Three of the ten runs stopped at the 30-epoch ceiling with validation loss still improving, so those three are lower bounds rather than converged results and reruns at a higher ceiling have been prepared. Grad-CAM analysis across all five models and the cross-generator evaluation on GenImage are the next steps.

Conclusion

The review paper is complete in draft: an abstract with six keywords, eight sections totalling roughly 9,850 words, a dataset comparison table covering 11 datasets, the master comparison table covering all 48 primary studies, and two figures. Every reference identifier in the bibliography — 74 in total — was opened and checked against its arXiv, Crossref or proceedings record, with the result logged one line per reference. The assembled draft was submitted for the departmental plagiarism check on 15 August 2026 and the result is awaited.

Three conclusions carry into the second phase. Generalisation across generators is a structural problem rather than an engineering inconvenience, because a change of generator design can remove the evidence a detector relies on rather than merely obscure it. The strongest published cross-generator results share one of three properties: a representation that was never trained on detection data, an input derived from the image rather than the image itself, or training data spanning many generators. And cross-generator evaluation and explanation analysis are almost never carried out together, which is precisely the gap this project addresses. What remains for Semester VII is restyling the report into the official Amity template, the plagiarism-check result and any corrections arising from it, the team's own rewrite pass over the drafted prose, and the guide review in the final week.

PERT Chart/Schedule of project completion:-

Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Scope, title finalisation, search protocol and section skeleton	✓											
Literature collection and title/abstract screening		✓										
Full-text reading of 48 primary studies; extraction table			✓									
Taxonomy finalisation and Section 1 (Introduction)				✓								
Section 2 (Background: GANs, diffusion models, artefacts)					✓							
Section 3 (Taxonomy of detection approaches, 3.1-3.5)						✓	✓					
Section 4 (Datasets) and Section 5 (Metrics) with tables and figures								✓				
Master comparison table (48 studies) and Section 6									✓			
Section 7 (Gap analysis), Section 8 (Conclusion) and Abstract										✓		
Report assembly, reference verification, plagiarism/AI-text check											✓	
Guide review, corrections, final formatting and submission												✓
Implementation environment, CIFAKE pipeline and split file				✓	✓							
Transfer-learning experiment matrix (5 backbones x 2 modes)					✓	✓						
Grad-CAM analysis and cross-generator evaluation						✓	✓	✓				

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